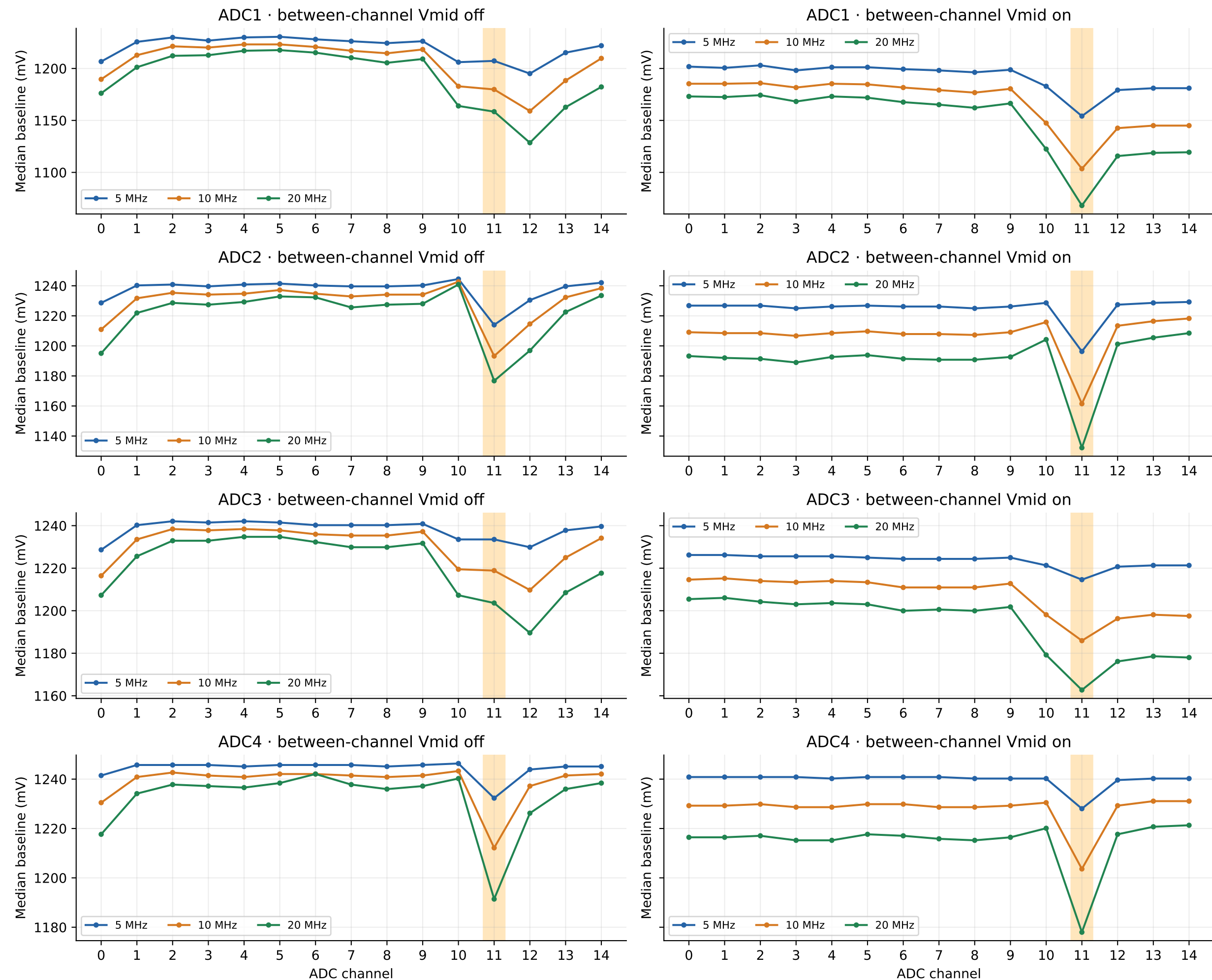
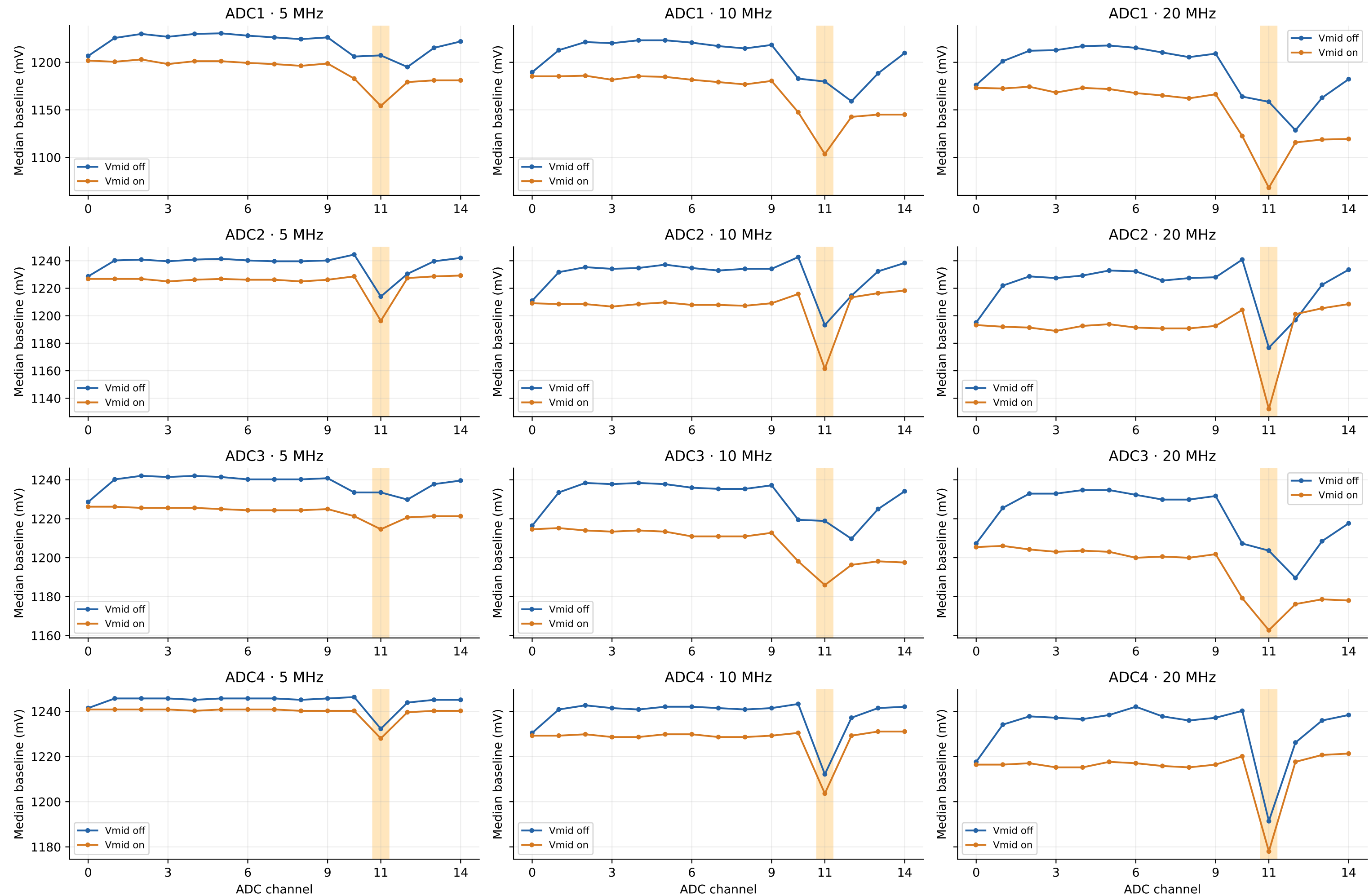


SPI speed comparison — all four ADCs, channels 0-14



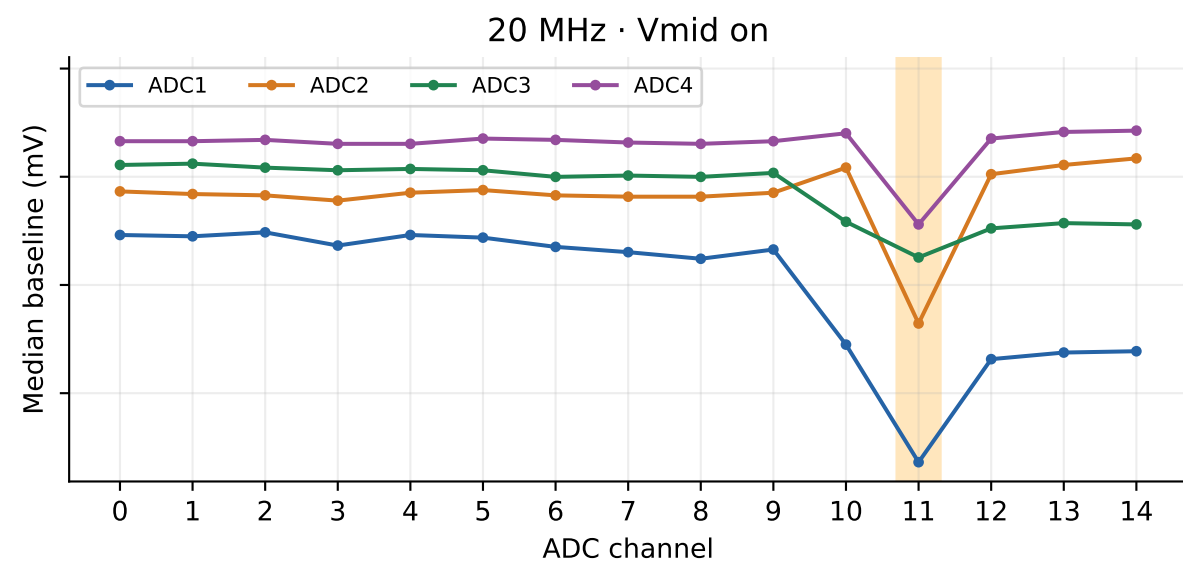
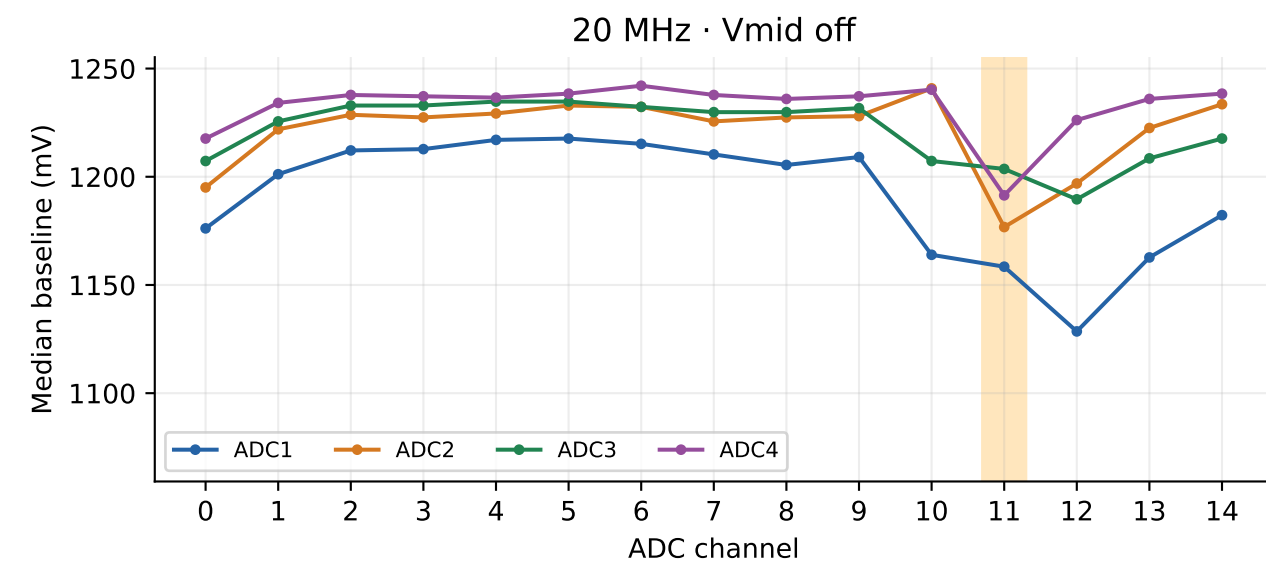
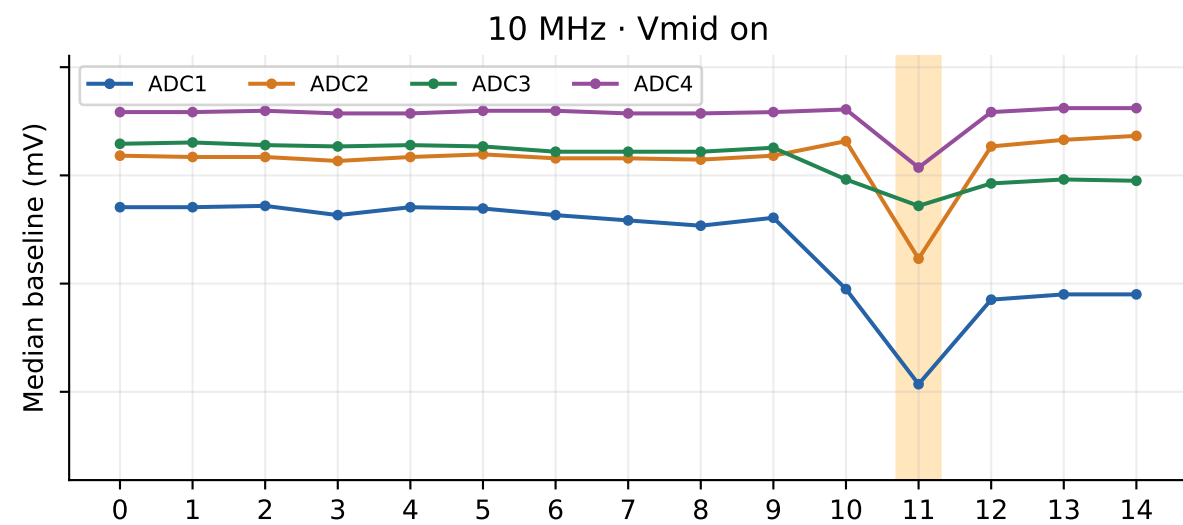
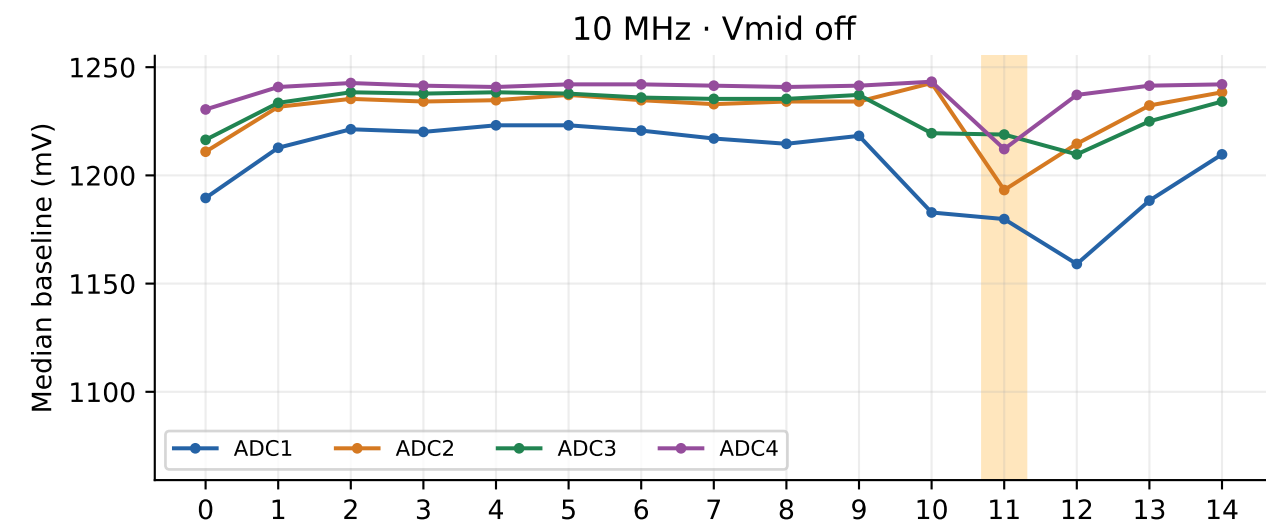
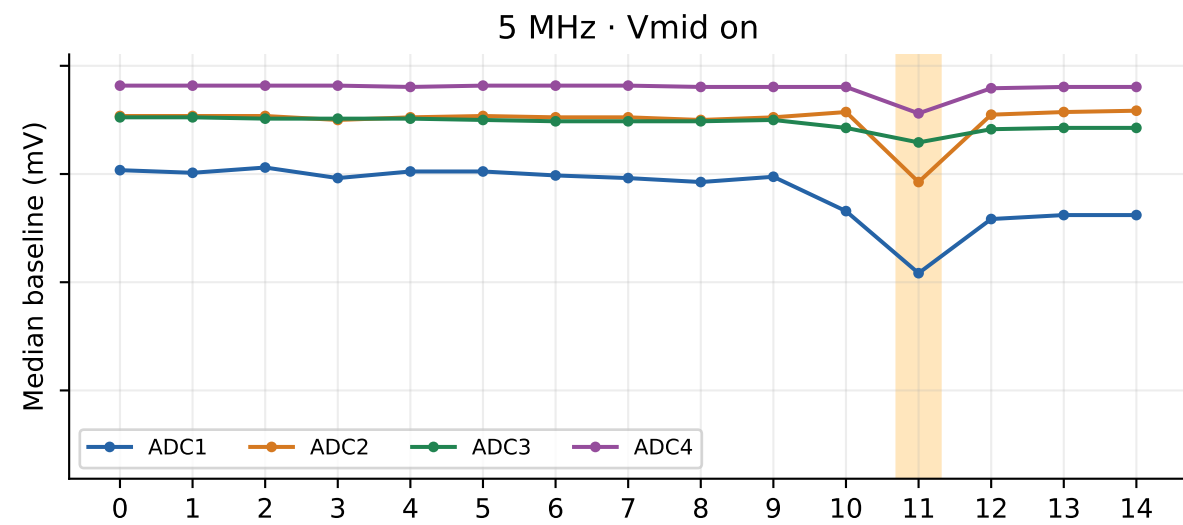
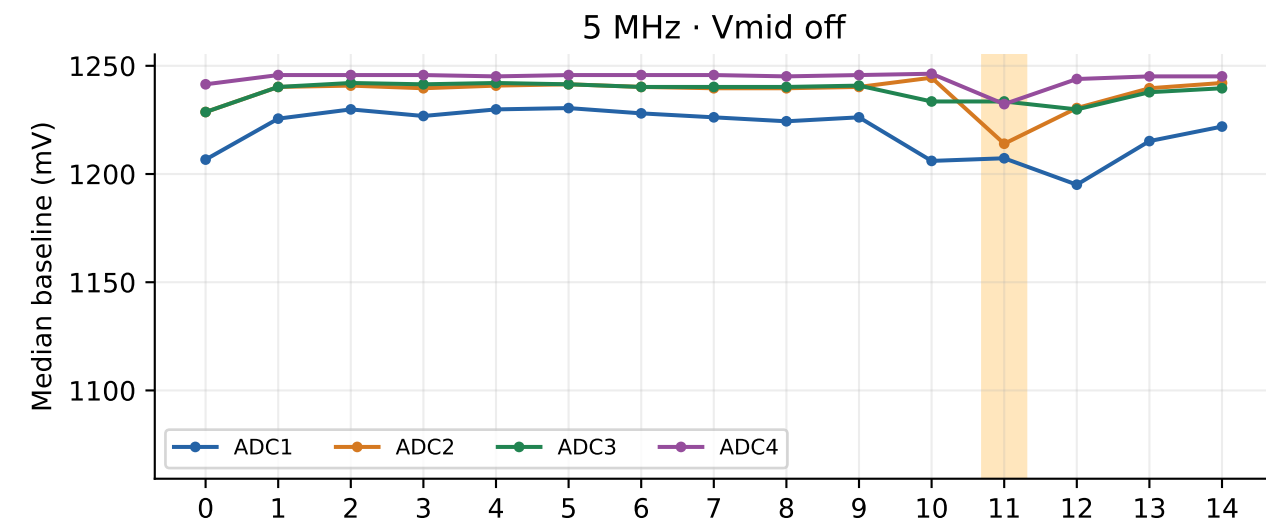
Exact 60-route scans; channel 11 highlighted. 12 s after 1 s warm-up; no connected sensor strips.

Effect of Vmid between channels — baseline versus channel



Exact 60-route scans; same manual/blocking/repeat1 settings. mV = raw code × 2500/4096.

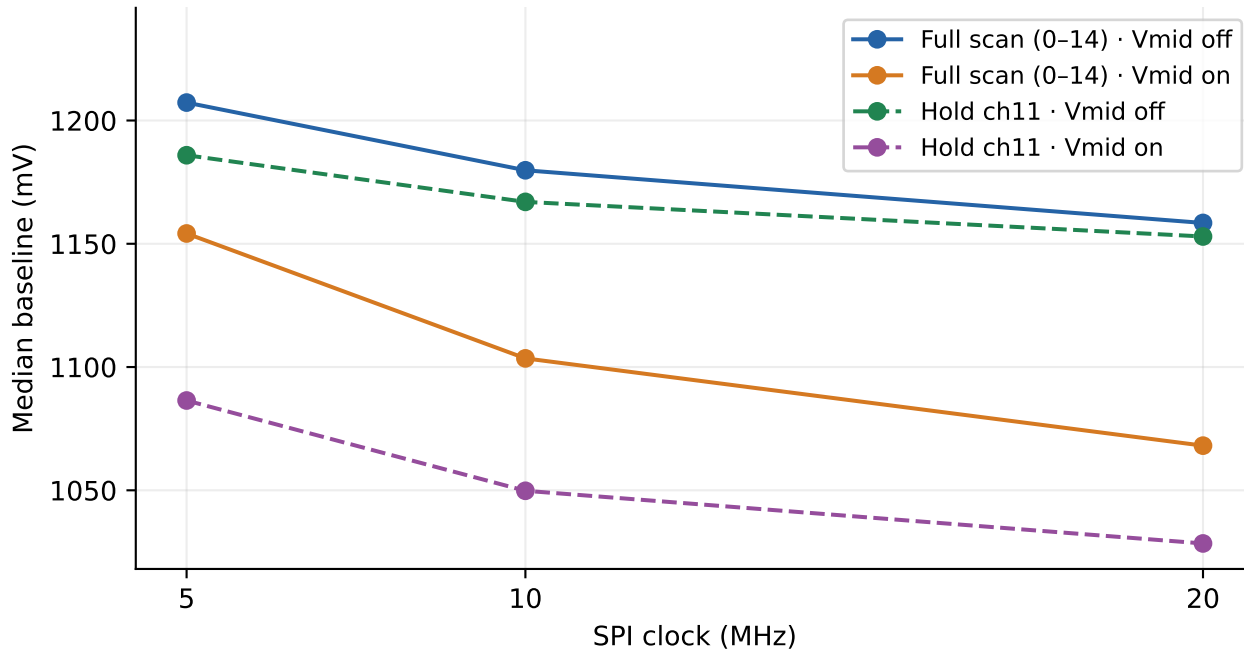
Four ADCs at the same SPI speed



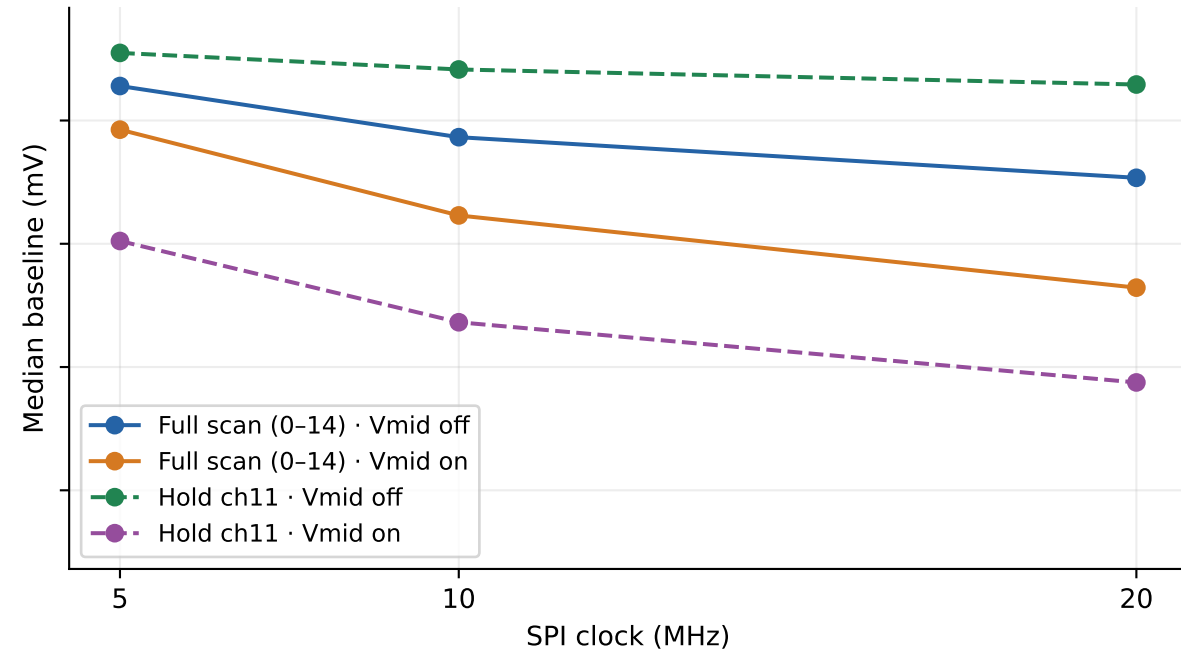
All four ADCs captured together, each with channels 0–14. Channel 11 highlighted; medians in mV.

Channel 11 — scanning versus holding one ADC alone

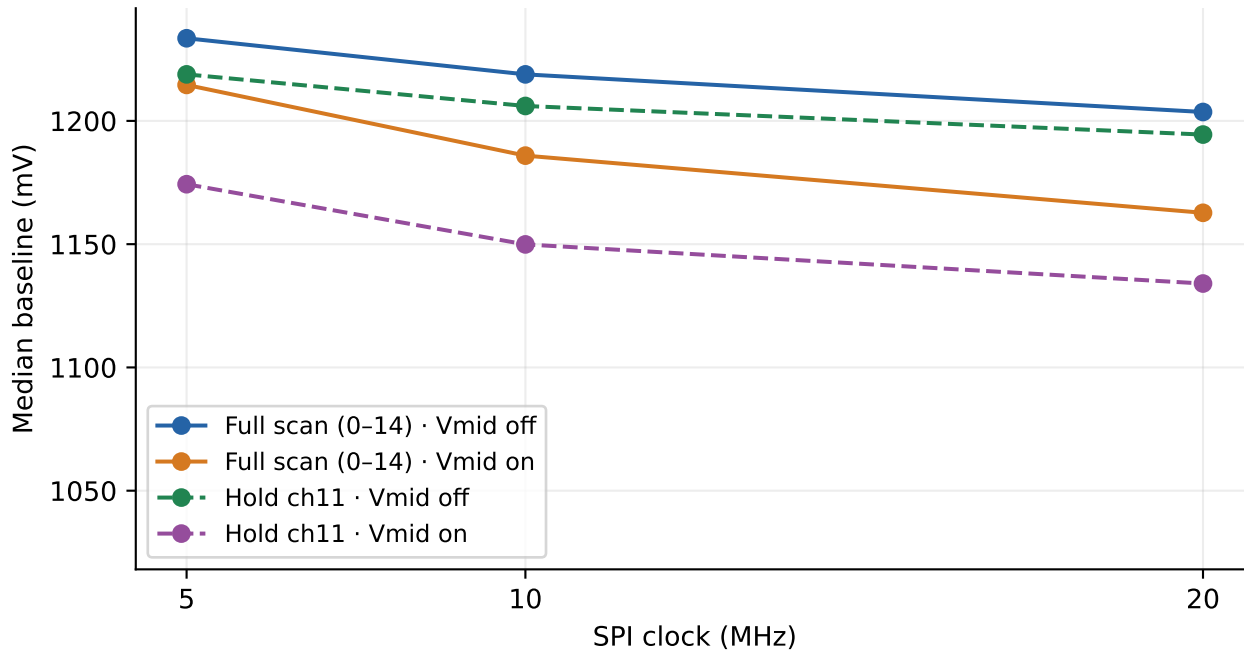
ADC1 · channel 11



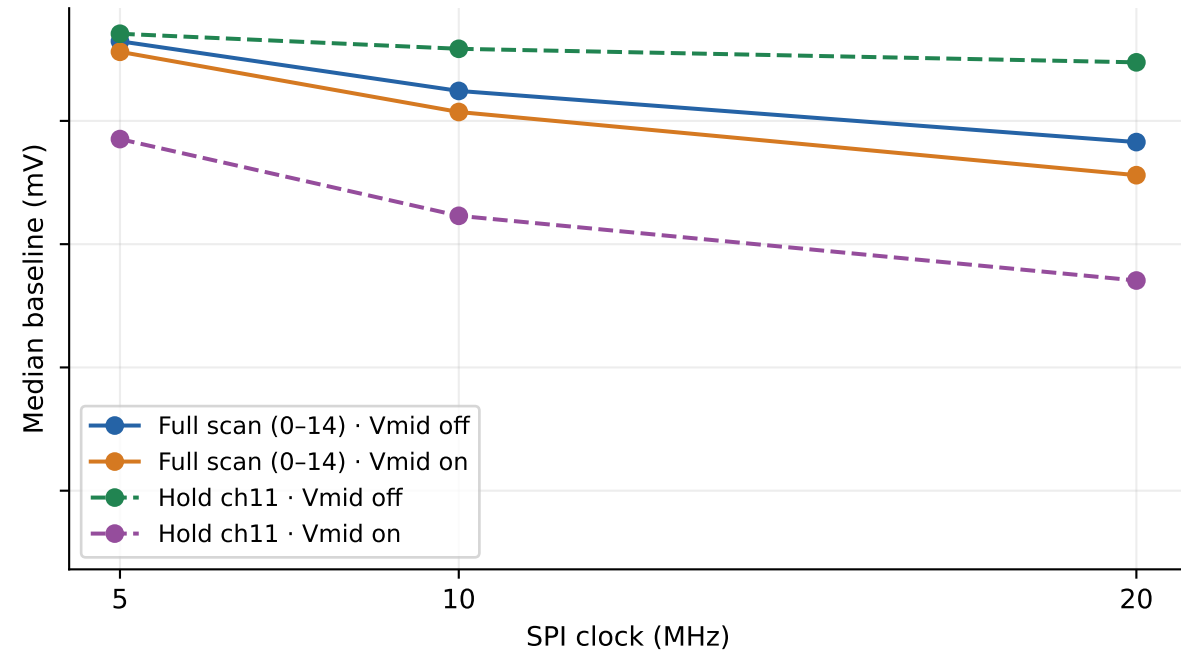
ADC2 · channel 11



ADC3 · channel 11

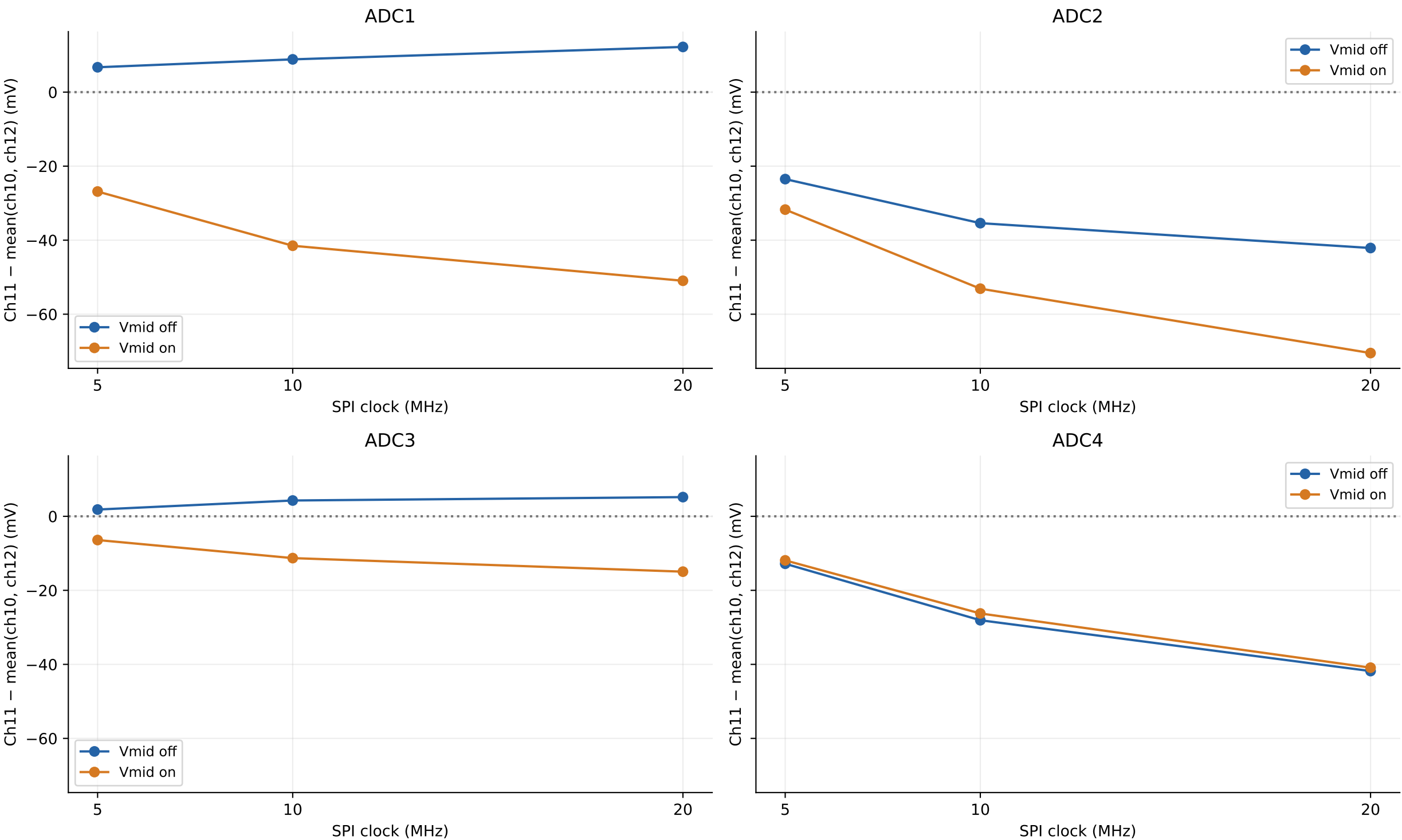


ADC4 · channel 11



Full scan: all 4 ADCs, channels 0-14; curves show only ch11. Hold: one ADC alone selects ch11 repeatedly.
Vmid-on hold repeats 11 → 15 → 15; only ch11 retained. At 10 MHz: scan off/on 159/415 μ s; either hold 8 μ s.

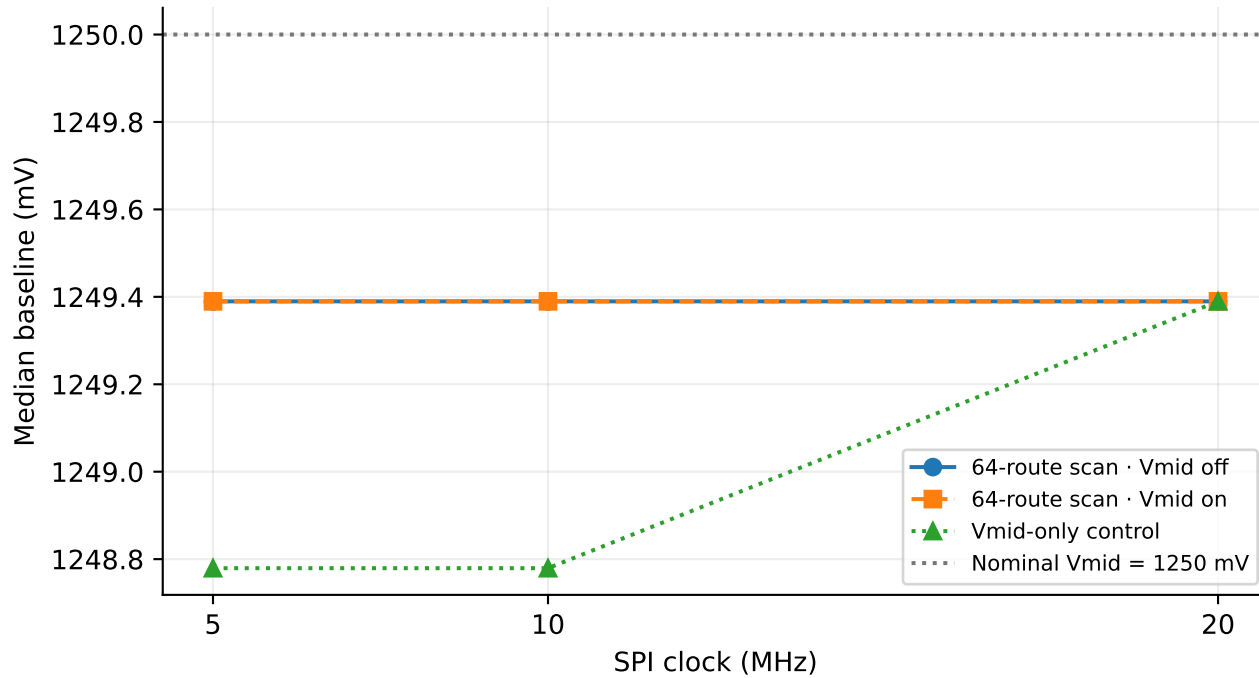
Channel 11 relative to its neighbors — full 60-route scans



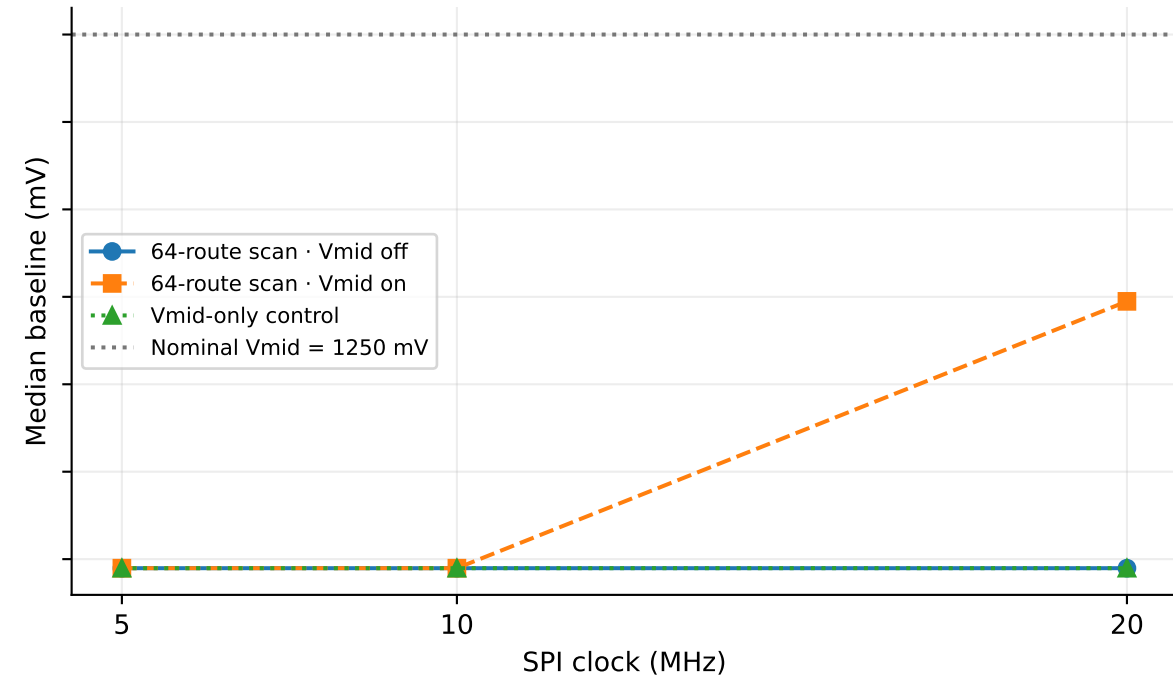
Negative values mean channel 11 reads below its neighbors. This removes part of the ADC-wide baseline shift.

Measured Vmid / channel 15 — all four ADCs

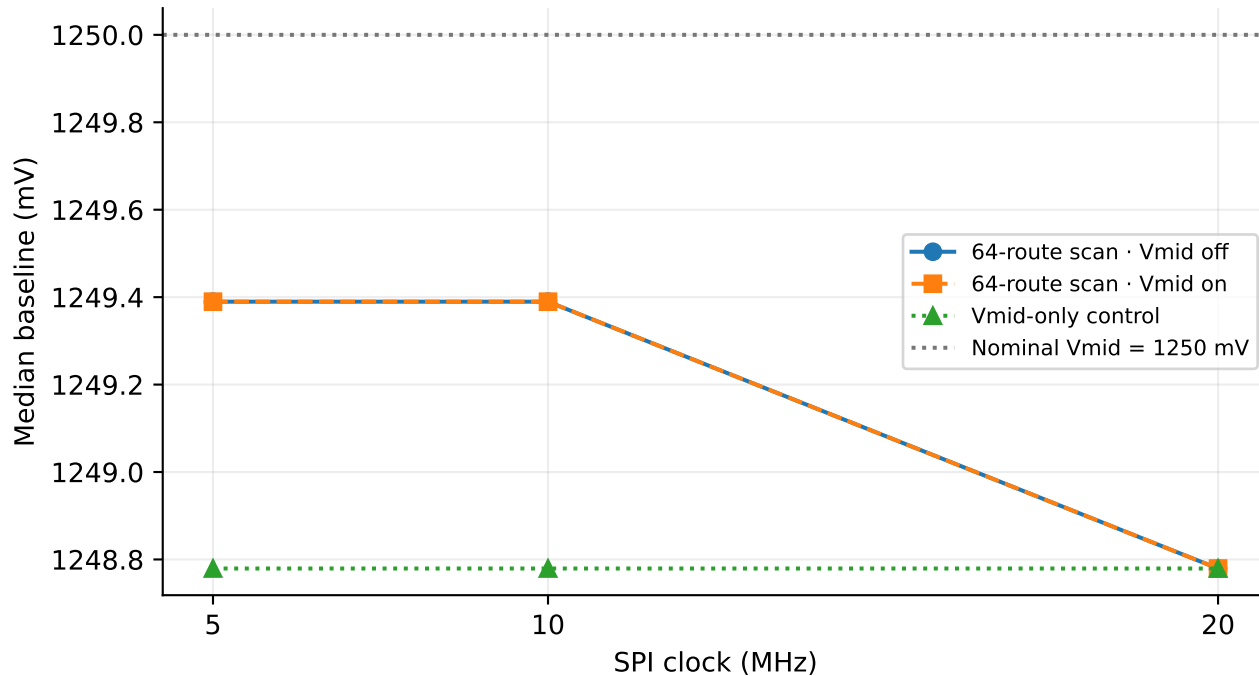
ADC1 · direct channel 15



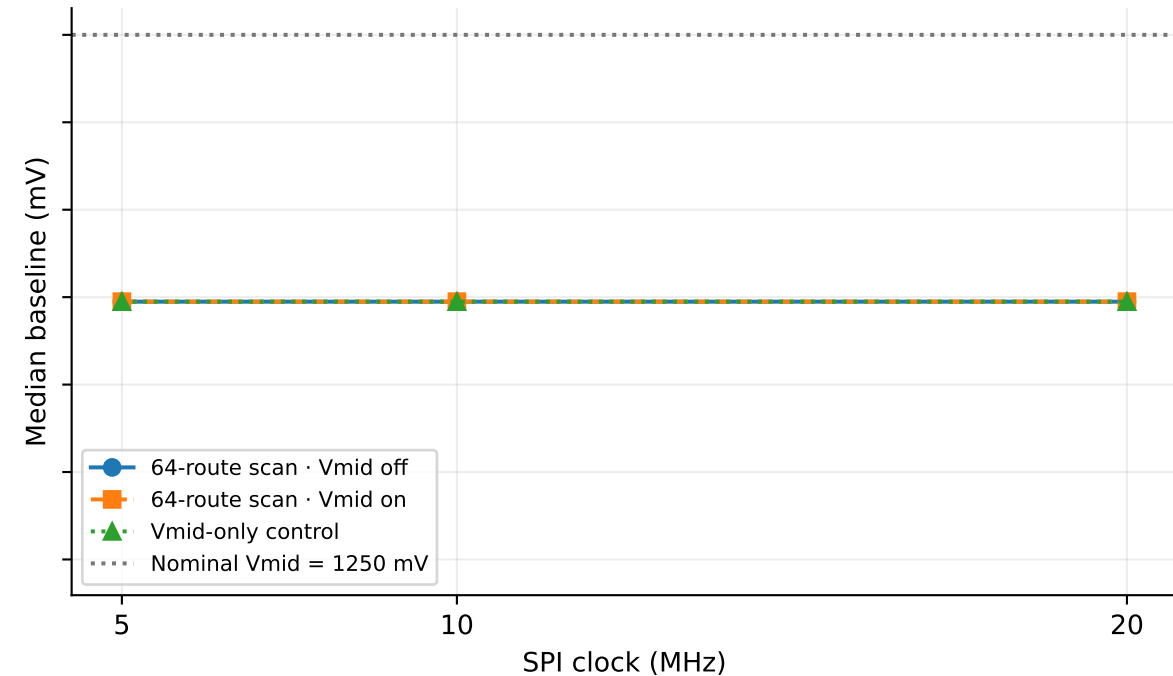
ADC2 · direct channel 15



ADC3 · direct channel 15

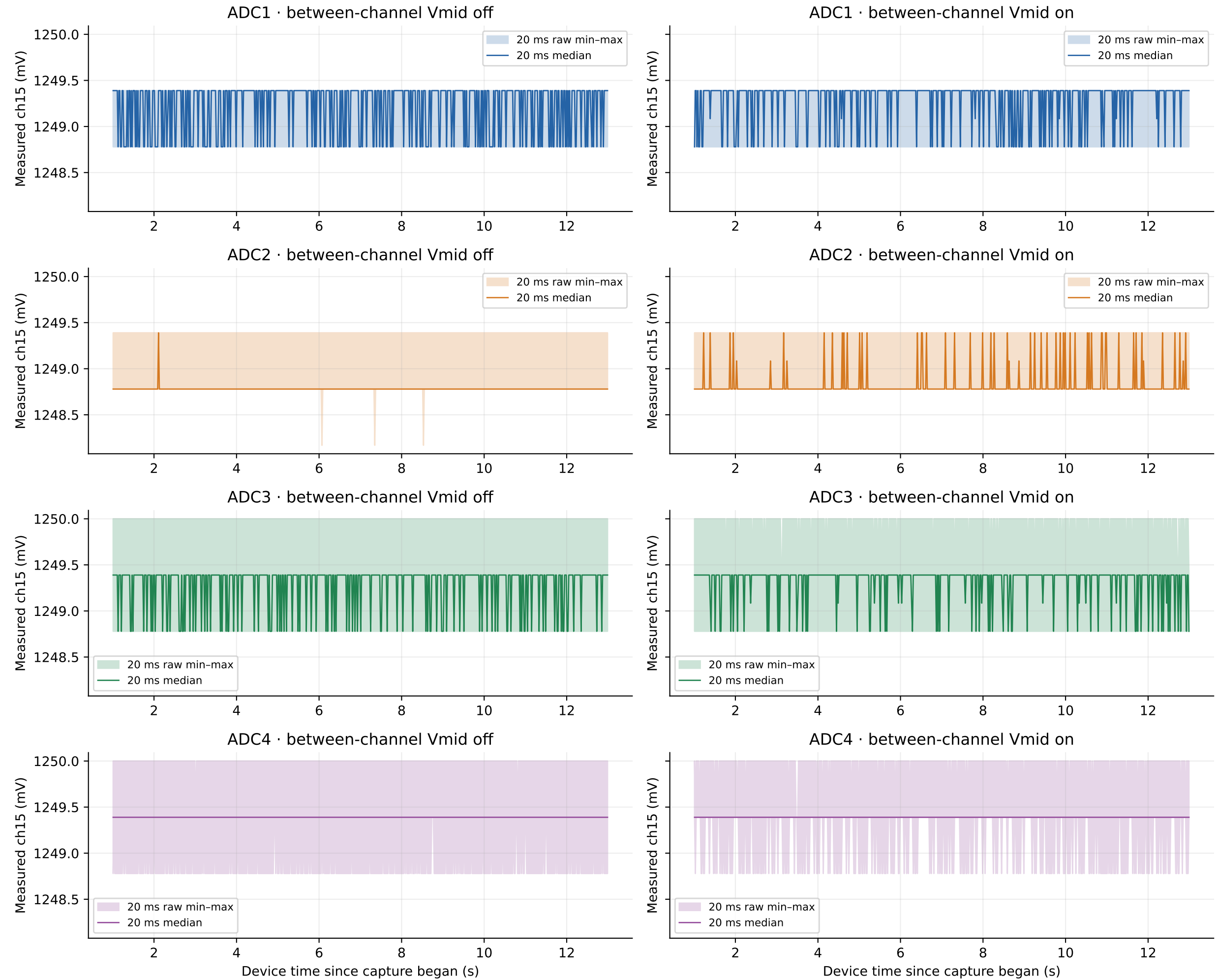


ADC4 · direct channel 15



Companion 64-route scans retain ch15 after ch14; reference-only controls are separate captures, not derived estimates.

Vmid time traces at 10 MHz — every sample retained in the envelope



Direct reference conversions from companion 64-route scans. Shaded envelope includes all samples; no interpolation or noise filtering.